

# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

## ASSESSMENT TOOL 1 – Data Interpretation

|                  |                                                                                                                                   |               |                                         |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------|---------------|-----------------------------------------|
| Course Code:     | DSA03                                                                                                                             | Course Title: | Natural Language Processing             |
| CO Assessed:     | CO4 – Precision                                                                                                                   | Assessment:   | Assessment Tool 1 – Data Interpretation |
| Weightage:       | 35% of CIA                                                                                                                        |               |                                         |
| CO5 Statement:   | Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4) |               |                                         |
| Student Name:    | G.Vinay Kumar Reddy                                                                                                               | Reg. No.:     | 192472093                               |
| Section / Batch: | CSE-AI                                                                                                                            | Date:         | 20/08/2026                              |

### Code of Conduct

*I G.Vinay Kumar Reddy (192472093) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: vinay

### Instructions

- Read each data set, table, semantic representation, or scenario carefully before answering.
- Analyze the given information and provide logical, evidence-based interpretations.
- Show all intermediate reasoning, semantic analysis steps, predicate representations, or calculations wherever applicable.
- Answers should be concise, structured, and technically accurate.
- Support your conclusions using the data provided in the question.
- Draw diagrams, semantic networks, parse trees, or logical representations wherever relevant.
- Avoid copying definitions alone; focus on analyzing the given scenario and interpreting the data.
- Duration: 30 minutes.

| Section / Topic                                                                | Focus                                                                                                                                                         | Question No. |
|--------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| Representing Meaning & Meaning Structure of Language                           | Analyze semantic representations, identify action–entity relationships, detect interpretation errors, and evaluate meaning representation in chatbot systems. | Q1           |
| First-Order Predicate Calculus & Representing Linguistically Relevant Concepts | Construct predicate representations, apply logical inference rules, derive conclusions, and evaluate reasoning in smart manufacturing environments.           | Q2           |
| Lexemes and Their Senses & Word Sense Disambiguation                           | Analyze contextual clues, determine correct word senses, evaluate ambiguity resolution, and improve semantic understanding in search systems.                 | Q3           |
| Syntax-Driven Semantic Analysis                                                | Analyze syntactic structures, assign semantic roles, evaluate parsing accuracy, and improve semantic interpretation in healthcare NLP applications.           | Q4           |

1. Frame Semantics in Banking Customer-Service Systems

A digital banking platform uses semantic frames to understand customer requests. The NLP system identifies the **action, agent, object, and destination** involved in each request.

| Customer Query                                                  | Generated Semantic Frame                                      |
|-----------------------------------------------------------------|---------------------------------------------------------------|
| "Transfer ₹20,000 from my savings account to my son's account." | TRANSFER(SourceAccount, Amount, DestinationAccount, Customer) |
| "Block my debit card immediately."                              | BLOCK(DebitCard, Customer)                                    |
| "Send my transaction statement to my email."                    | SEND(TransactionStatement, Email, Customer)                   |
| "Increase my daily withdrawal limit."                           | MODIFY(WithdrawalLimit, Customer)                             |

Additional transaction logs show:

| Query ID | Actual User Intent                     | System Interpretation     |
|----------|----------------------------------------|---------------------------|
| B1       | Transfer money to another account      | Transfer money            |
| B2       | Block debit card                       | Block debit card          |
| B3       | Receive transaction statement by email | Send statement to email   |
| B4       | Increase withdrawal limit              | Decrease withdrawal limit |

Tasks:

1. Analyze the semantic frame of each banking request and identify the relationship between the action and its participants.
2. Identify the query in which the semantic interpretation is incorrect and explain why.
3. Analyze how incorrect identification of semantic roles can lead to an incorrect banking operation.
4. Recommend a frame-based semantic analysis approach to improve the reliability of banking conversational systems.

2. First-Order Predicate Calculus for Smart Agriculture

A smart agricultural system monitors crop fields using sensors and logical rules.

Field Data:

| Field | Soil Condition | Irrigation Status | Crop  |
|-------|----------------|-------------------|-------|
| F1    | Dry            | Available         | Rice  |
| F2    | Wet            | Available         | Wheat |
| F3    | Dry            | Unavailable       | Maize |
| F4    | Wet            | Unavailable       | Rice  |

The system uses the following predicate rules:

- $Dry(x) \wedge IrrigationAvailable(x) \rightarrow NeedsWater(x)$
- $NeedsWater(x) \rightarrow Irrigate(x)$
- $Wet(x) \rightarrow \neg NeedsWater(x)$
- $IrrigationUnavailable(x) \rightarrow \neg Irrigate(x)$
- $Rice(x) \wedge NeedsWater(x) \rightarrow PriorityIrrigation(x)$

Tasks:

1. Represent the field observations using First-Order Predicate Calculus.
2. Using the given rules, determine which fields require irrigation.
3. Determine whether F1 and F3 should be treated differently by the automated irrigation controller.
4. Analyze whether predicate logic alone is sufficient for agricultural decision support when sensor information is incomplete or uncertain.

3. Word Sense Disambiguation in a Travel Recommendation System

A travel platform receives search queries containing ambiguous words. The recommendation engine uses surrounding words and user interactions to determine the intended meaning.

| Search Query     | Possible Sense 1  | Possible Sense 2          |
|------------------|-------------------|---------------------------|
| "Amazon tour"    | Amazon rainforest | Amazon company            |
| "Safari booking" | Wildlife tour     | Web browser               |
| "Java vacation"  | Java island       | Java programming language |
| "Cruise package" | Ship journey      | Cruise software/project   |

The system records the following user interactions:

| Query          | User Interaction                    |
|----------------|-------------------------------------|
| Amazon tour    | Viewed rainforest trekking packages |
| Safari booking | Selected wildlife resort packages   |
| Java vacation  | Viewed hotels in Bali and Java      |
| Cruise package | Viewed Mediterranean ship packages  |

However, another user searches: **"Safari download for Windows"** and clicks a browser download page.

#### Tasks:

1. Determine the intended sense of the ambiguous terms using the available context.
2. Identify the lexical and contextual cues that distinguish the possible senses.
3. Analyze why the same word may require different interpretations for different users.
4. Design an industrial-scale WSD strategy using query context, user history, embeddings, and click behaviour to improve travel recommendations.

#### 4. Syntax-Driven Semantic Analysis in Legal Document Processing

A legal NLP system analyzes sentences from contracts and assigns semantic roles based on their syntactic structures.

The system produces the following analyses:

| Legal Sentence                                             | Parse Structure      | Generated Semantic Interpretation                       |
|------------------------------------------------------------|----------------------|---------------------------------------------------------|
| "The supplier delivered the equipment to the buyer."       | Subject–Verb–Object  | Supplier = Agent, Equipment = Object, Buyer = Recipient |
| "The buyer received the equipment from the supplier."      | Subject–Verb–Object  | Buyer = Agent, Equipment = Object, Supplier = Recipient |
| "The company terminated the contract after the violation." | Subject–Verb–Object  | Company = Agent, Contract = Object                      |
| "The contract was terminated by the company."              | Passive Construction | Contract = Agent, Company = Object                      |

#### Tasks:

1. Analyze how syntactic structure, including active and passive constructions, influences semantic-role assignment.
2. Identify the incorrect semantic-role assignments in the given interpretations.
3. Explain how confusing grammatical subject with semantic agent can affect automated legal information extraction.
4. Recommend a syntax-driven semantic analysis architecture that can correctly handle active voice, passive voice, prepositional phrases, and long-distance dependencies in legal documents.

#### Rubrics for Calculation

| Criteria                | Weightage | Level 4 (Exemplary)                                                       | Level 3 (Proficient)                                 | Level 2 (Developing)                     | Level 1 (Beginning)                       |
|-------------------------|-----------|---------------------------------------------------------------------------|------------------------------------------------------|------------------------------------------|-------------------------------------------|
| Understanding of Data   | 25%       | Accurately interprets all data patterns, trends, and relationships        | Interprets data correctly with minor gaps            | Partial understanding; misses key trends | Misinterprets or fails to understand data |
| Analysis                | 25%       | Provides deep analysis with strong logical reasoning and justification    | Provides reasonable analysis with some justification | Basic analysis with weak reasoning       | No meaningful analysis provided           |
| Application of Concepts | 20%       | Effectively applies relevant concepts/tools to interpret data accurately  | Applies concepts with minor errors                   | Limited application; requires guidance   | Unable to apply concepts to data          |
| Conclusion              | 20%       | Draws clear, meaningful conclusions with strong insights and implications | Provides valid conclusions with moderate insights    | Conclusions are vague or weak            | No clear or valid conclusions             |
| Clarity                 | 10%       | Presents interpretation clearly using proper                              | Generally clear with minor issues                    | Lacks clarity or proper structure        | Poor presentation                         |

|  |  |                                     |  |  |                             |
|--|--|-------------------------------------|--|--|-----------------------------|
|  |  | structure, visuals, and terminology |  |  | and difficult to understand |
|--|--|-------------------------------------|--|--|-----------------------------|

| Q. No. / Criteria Level | Understanding of Data | Analysis | Application of Concepts | Conclusion | Clarity | Sub. Total | Total % |
|-------------------------|-----------------------|----------|-------------------------|------------|---------|------------|---------|
| 1                       | 3                     | 3        | 3                       | 4          | 3       | 17         | 78      |
| 2                       | 3                     | 3        | 3                       | 4          | 3       | 17         |         |
| 3                       | 3                     | 3        | 3                       | 4          | 3       | 16         |         |
| 4                       | 3                     | 3        | 3                       | 4          | 3       | 16         |         |

| Faculty In-charge     | Course Coordinator | Program Director |
|-----------------------|--------------------|------------------|
| Dr. Kumaragurubaran T |                    |                  |
| Signature: _____      | Signature: _____   | Signature: _____ |
| Date: _____           | Date: _____        | Date: _____      |

## 1. Frame Semantics in Banking Customer-Service Systems

### 1. Semantic frame analysis

A **semantic frame** represents an action or event together with its participants and their roles.

| Query                                                           | Frame                                                         | Action   | Participants and Roles                                                                        |
|-----------------------------------------------------------------|---------------------------------------------------------------|----------|-----------------------------------------------------------------------------------------------|
| “Transfer ₹20,000 from my savings account to my son's account.” | TRANSFER(SourceAccount, Amount, DestinationAccount, Customer) | Transfer | Customer = initiator, Savings Account = source, ₹20,000 = amount, Son's Account = destination |
| “Block my debit card immediately.”                              | BLOCK(DebitCard, Customer)                                    | Block    | Customer = requester, Debit Card = object                                                     |
| “Send my transaction statement to my email.”                    | SEND(TransactionStatement, Email, Customer)                   | Send     | Customer = requester, Transaction Statement = object, Email = destination                     |
| “Increase my daily withdrawal limit.”                           | MODIFY(WithdrawalLimit, Customer)                             | Modify   | Customer = requester, Withdrawal Limit = object, Increase = requested modification            |

The frame identifies **what action is requested and which entities participate in that action**.

### 2. Incorrect semantic interpretation

**B4 is incorrect.**

- Actual intent: **Increase withdrawal limit**
- System interpretation: **Decrease withdrawal limit**

The system correctly identified the object, *withdrawal limit*, but incorrectly interpreted the action direction. The word “**increase**” explicitly indicates that the value should become larger.

B1, B2, and B3 have interpretations consistent with their actual intents.

### 3. Effect of incorrect semantic roles

Incorrect semantic-role identification can result in a **wrong financial operation**.

For example, if:

“Increase my daily withdrawal limit”

is interpreted as:

“Decrease my daily withdrawal limit”

the banking system may actually reduce the customer's permitted withdrawal amount.

Similarly, incorrectly identifying the source and destination accounts in a transfer could potentially result in money being transferred to the wrong account. Therefore, semantic errors in banking systems can have serious financial and security consequences.

#### 4. Recommended frame-based approach

A reliable banking conversational system should use a **frame-based semantic pipeline**:

**User Query → Intent Detection → Entity Recognition → Semantic Role Assignment → Frame Construction → Validation → Banking Operation**

The system should:

1. Detect the main action such as TRANSFER, BLOCK, SEND, or MODIFY.
2. Extract entities such as account numbers, amounts, cards, and email addresses.
3. Assign semantic roles such as **source, destination, object, amount, customer, and recipient**.
4. Detect modifiers such as **increase/decrease, immediately, and recurring**.
5. Validate the complete frame against banking rules before executing an operation.
6. Request confirmation for high-risk transactions.

This approach improves **accuracy, explainability, and transaction safety**.

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## 2. First-Order Predicate Calculus for Smart Agriculture

### 1. Representing observations using FOPC

The field observations can be represented using predicates:

- Dry(x) — Field x has dry soil.
- Wet(x) — Field x has wet soil.
- IrrigationAvailable(x) — irrigation is available for field x.
- IrrigationUnavailable(x) — irrigation is unavailable.
- Rice(x) — field x contains rice.
- Wheat(x) — field x contains wheat.
- Maize(x) — field x contains maize.

The observations are:

**F1**

- Dry(F1)
- IrrigationAvailable(F1)
- Rice(F1)

**F2**

- Wet(F2)
- IrrigationAvailable(F2)
- Wheat(F2)

**F3**

- Dry(F3)
- IrrigationUnavailable(F3)
- Maize(F3)

#### F4

- Wet(F4)
- IrrigationUnavailable(F4)
- Rice(F4)

The rules are:

## 2. Fields requiring irrigation

#### F1

Given:

Dry(F1) and IrrigationAvailable(F1)

Therefore:

NeedsWater(F1)

Then:

Irrigate(F1)

Since F1 contains rice:

PriorityIrrigation(F1)

**F1 requires irrigation and has priority.**

#### F2

Wet(F2) gives:

$\neg$ NeedsWater(F2)

Therefore, **F2 does not require irrigation.**

#### F3

Dry(F3) suggests that it needs water, but irrigation is unavailable.

The first rule cannot be applied because:

IrrigationAvailable(F3) is false.

Therefore, NeedsWater(F3) cannot be derived from the supplied rules.

Also:

$\text{IrrigationUnavailable(F3)} \rightarrow \neg \text{Irrigate(F3)}$

Hence, **F3 cannot be irrigated automatically.**

#### F4

$\text{Wet(F4)} \rightarrow \neg \text{NeedsWater(F4)}$

Therefore, **F4 does not require irrigation.**

## Final result

| Field | Needs Water   | Irrigation Possible? | Controller Action |
|-------|---------------|----------------------|-------------------|
| F1    | Yes           | Yes                  | Irrigate          |
| F2    | No            | Yes                  | Do not irrigate   |
| F3    | Not derivable | No                   | Do not irrigate   |
| F4    | No            | No                   | Do not irrigate   |

### 3. Should F1 and F3 be treated differently?

Yes.

Both F1 and F3 have **dry soil**, but their irrigation availability differs.

- **F1:** Dry + irrigation available → irrigation should be activated.
- **F3:** Dry + irrigation unavailable → irrigation cannot be activated.

Therefore, the controller should not treat soil condition alone as sufficient. It must also consider **resource availability**.

F1 receives irrigation, while F3 may require an alternative action such as notifying the farmer or activating another available water source.

### 4. Is predicate logic sufficient?

**No. Predicate logic alone is not sufficient for real-world agricultural decision support.**

FOPC works well with clear facts, such as:

Dry(F1)

However, real sensors may provide uncertain or incomplete information:

- Soil moisture sensor may malfunction.
- A sensor may report 45% moisture with uncertainty.
- Weather forecasts may be unreliable.
- Irrigation availability may change.
- Different sensors may provide conflicting measurements.

Classical predicate logic generally represents facts as **true or false**, whereas agricultural systems often need **probabilities, confidence values, fuzzy values, and temporal information**.

A practical system could combine FOPC with:

- **Fuzzy logic** for gradual soil-moisture conditions.
- **Probabilistic reasoning** for uncertain sensor data.
- **Machine learning** for prediction.
- **Temporal reasoning** for changing weather and soil conditions.

Thus, predicate logic is useful as a rule-based foundation but should be combined with uncertainty-handling techniques.

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## 3. Word Sense Disambiguation in a Travel Recommendation System

### 1. Determining the intended senses

| Query                       | Intended Sense                 | Evidence                                 |
|-----------------------------|--------------------------------|------------------------------------------|
| Amazon tour                 | Amazon rainforest              | User viewed rainforest trekking packages |
| Safari booking              | Wildlife tour                  | User selected wildlife resort packages   |
| Java vacation               | Java island/travel destination | User viewed hotels in Bali and Java      |
| Cruise package              | Ship journey                   | User viewed Mediterranean ship packages  |
| Safari download for Windows | Web browser                    | User clicked a browser download page     |

The system should use both the query and subsequent user behaviour to determine the intended meaning.



## 2. Lexical and contextual cues

### Amazon

- **Tour, trekking, rainforest, packages** → Amazon rainforest
- **Shopping, AWS, cloud, Prime** → Amazon company

### Safari

- **Booking, wildlife, resort, animals** → wildlife safari
- **Download, Windows, browser, Apple** → Safari web browser

### Java

- **Vacation, hotels, Bali, Indonesia, tourism** → Java island
- **Programming, code, JVM, developer** → Java programming language

### Cruise

- **Mediterranean, ship, cabin, package, travel** → cruise journey
- **software, project, development** → software/project named Cruise

Thus, the meaning is determined by the **surrounding lexical context and user behaviour**, rather than the ambiguous word alone.

## 3. Why interpretation differs between users

The same word can have different meanings depending on:

1. **Query context**
2. **Previous searches**
3. **User interests**
4. **Geographical context**
5. **Click behaviour**
6. **Browsing history**
7. **Current task or session**

For example:

“Safari booking”

strongly suggests a wildlife trip.

But:

“Safari download for Windows”

strongly indicates the browser.

Therefore, WSD should be **context-sensitive and user-sensitive** rather than assigning one permanent meaning to a word.

## 4. Industrial-scale WSD strategy

A practical WSD architecture can be designed as:

**Query → Context Extraction → Candidate Senses → Embedding Model → User History → Click Behaviour → Sense Ranking → Recommendation**

### Step 1: Query processing

Extract:

- Keywords
- Named entities
- Surrounding words
- Query intent
- Location information

### Step 2: Generate candidate senses

For **Java**, generate:

- Java island
- Java programming language

### Step 3: Context embeddings

Use Transformer-based embeddings to represent the query and compare it with representations of candidate senses.

For example:

“Java vacation hotels”

will have greater semantic similarity to **Java island** than Java programming.

### Step 4: User-history features

Consider previous searches and viewed products. If a user frequently searches for hotels and flights, a travel-related interpretation should receive a higher score.

### Step 5: Click behaviour

Clicks provide strong behavioural evidence.

For example:

“Safari download for Windows” → browser download clicked  
should increase the probability of the **web browser** sense.

### Step 6: Sense-ranking model

A ranking model can combine:

The sense with the highest score is selected.

### Step 7: Continuous learning

The system can learn from:

- Click-through rates
- Booking conversions
- Search reformulations
- User feedback
- Abandoned recommendations

This makes the WSD system scalable and adaptive for large travel platforms.

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## 4. Syntax-Driven Semantic Analysis in Legal Document Processing

### 1. Effect of syntactic structure on semantic roles

Syntactic structure provides important clues for assigning semantic roles, but **grammatical position and semantic role are not always identical**.

#### Active voice

“The supplier delivered the equipment to the buyer.”

- Subject = supplier
- Verb = delivered
- Object = equipment
- PP recipient = buyer

Semantic roles:

- Supplier = **Agent**
- Equipment = **Object/Theme**
- Buyer = **Recipient**

#### Another active sentence

“The buyer received the equipment from the supplier.”

Here:

- Buyer = grammatical subject
- Equipment = object
- Supplier = introduced through from

The verb **receive** changes the mapping of grammatical roles to semantic roles. Buyer is the **Recipient**, while supplier is the **Source/Agent**.

#### Passive voice

“The contract was terminated by the company.”

Although **contract** is the grammatical subject, it is not the semantic agent.

- Contract = **Patient/Theme**
- Company = **Agent**

The phrase **“by the company”** identifies the actual agent.

### 2. Incorrect semantic-role assignments

There are two important errors.

#### Error 1

“The buyer received the equipment from the supplier.”

The given interpretation says:

Buyer = Agent, Equipment = Object, Supplier = Recipient

This is incorrect.

A better interpretation is:

- **Buyer = Recipient**
- **Equipment = Theme/Object**
- **Supplier = Source/Agent**

The supplier provides the equipment, while the buyer receives it.

## Error 2

“The contract was terminated by the company.”

The given interpretation says:

Contract = Agent, Company = Object

This is incorrect.

Correct interpretation:

- **Contract = Patient/Theme**
- **Company = Agent**

The contract undergoes the termination, while the company performs the action.

The interpretations for the first and third sentences are broadly correct.

## 3. Problems caused by confusing grammatical subject with semantic agent

A major problem in legal NLP is assuming:

**Subject = Agent**

This is not always true.

Consider:

“The contract was terminated by the company.”

The grammatical subject is **contract**, but the semantic agent is **company**.

If an automated legal system incorrectly labels the contract as the agent, it could extract the wrong information about **who performed the legal action**.

This can affect:

- Contract analysis
- Liability identification
- Responsibility extraction
- Legal event extraction
- Compliance analysis
- Case-law analysis
- Automated question answering

For example, a system might incorrectly conclude:

“The contract terminated the company.”

instead of:

“The company terminated the contract.”

Such an error could seriously distort the meaning of a legal document.

## 4. Recommended syntax-driven semantic architecture

A robust legal NLP system should use a multi-stage architecture:

**Legal Text → Tokenization → Dependency/Constituency Parsing → Syntactic Analysis → Semantic Role Labeling → Coreference Resolution → Long-Distance Dependency Handling → Legal Event Extraction**

## **Module 1: Syntactic parsing**

Use dependency or constituency parsing to identify:

- Subject
- Verb
- Object
- Complements
- Prepositional phrases

## **Module 2: Voice detection**

Determine whether the sentence is:

- Active
- Passive
- Causative
- Nominalized

For passive sentences, the system should specifically inspect **by-phrases** to identify potential agents.

## **Module 3: Semantic Role Labeling**

Map syntactic constituents to semantic roles:

- Agent
- Patient/Theme
- Recipient
- Source
- Instrument
- Location
- Time

## **Module 4: Prepositional-phrase analysis**

Different prepositions can indicate different roles.

For example:

“to the buyer” → Recipient

“from the supplier” → Source

“by the company” → Agent in passive construction

## **Module 5: Coreference resolution**

Resolve references such as:

“The supplier delivered the equipment. **It** was inspected by the buyer.”

The system must determine what “**it**” refers to.

## **Module 6: Long-distance dependency handling**

Legal sentences can contain deeply embedded clauses:

“The company that the supplier appointed delivered the equipment that the buyer requested.”

Dependency parsing and Transformer-based models can help connect related words across long distances.

### **Recommended architecture**

A hybrid syntax-driven and neural semantic architecture would be most effective:

**Dependency Parser + Semantic Role Labeler + Coreference Resolver + Transformer Model + Legal Ontology**

This combines the structural interpretability of syntactic parsing with the contextual understanding of modern neural NLP models. It is particularly suitable for complex legal documents containing **active/passive constructions, prepositional phrases, nested clauses, and long-distance dependencies.**